Engineering Mathematics III

ENGE702

Lecture 7: Second Order Homogeneous ODEs

Second order ODEs

In this lecture we will look at methods of solution for a certain class of second order ODEs. Specifically, we will look at **second order linear homogeneous ODEs with constant coefficients**. These are ODEs of the form

$$y'' + ay' + by = 0$$

where a and b are constants.

The form above is 'standard form', where the coefficient in front of the y'' is 1. If we have a second order linear homogeneous ODEs with constant coefficients that is not in standard form, we can always put it in standard form by dividing through by whatever is in front of the y'' (this is never 0, otherwise it would not be second order).

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The method to solve these is quite simple, and is based on our observation from an earlier lecture that the solution equation

$$y' + ky = 0 \quad \text{is} \quad y = e^{-kx}.$$

Since this essentially is the first order version of our second order equation, it may be reasonable to 'assume' this will be a soluton for our second order equation as well, and see what happens. Lets assume that the solution of

$$y'' + ay' + by = 0$$

is

$$y = e^{\lambda x}$$

for some constant λ. Then

$$y' = \lambda e^{\lambda x} \quad \text{and} \quad y'' = \lambda^2 e^{\lambda x}.$$

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Putting these values into our original equation $y'' + ay' + by = 0$, we have

$$(\lambda^2 + a\lambda + b)e^{\lambda x} = 0.$$

So if we can find a solution to this equation (ie find an appropriate λ to make it work), we have found a solution to our second order differential equation.

Since $e^{\lambda x}$ is never 0, it must mean that $\lambda^2 + a\lambda + b = 0$. This is simply solving a quadratic polynomial!

This equation $\lambda^2 + a\lambda + b = 0$ is called the **characteristic equation** of its corresponding differential equation.

We can do this with the quadratic formula (if necessary), to get solutions

$$\lambda_1 = \frac{1}{2}\left(-a + \sqrt{a^2 - 4b}\right) \quad \text{and} \quad \lambda_2 = \frac{1}{2}\left(-a - \sqrt{a^2 - 4b}\right)$$

giving solutions

$$y_1 = e^{\lambda_1 x} \quad \text{and} \quad y_2 = e^{\lambda_2 x}$$

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Remember that we need two linearly independent solutions for a general solution. We have two solutions above, but these will only give non-proportional real valued solutions if $a^2 - 4b > 0$. In other cases we might have to do some more work. We will show how to get solutions for all possible cases.

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Case 1: Two real roots (ie $a^2 - 4b > 0$):

In this case the solutions

$$y_1 = e^{\lambda_1 x} \quad \text{and} \quad y_2 = e^{\lambda_2 x}$$

are linearly independent, so we have the general solution:

$$y = c_1 e^{\lambda_1 x} + c_2 e^{\lambda_2 x}$$

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Example: Find the general solution to the differential equation

$$y'' + y' - 2y = 0$$

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Example: Find the general solution to the differential equation

$$y'' + y' - 2y = 0$$

This is a second order linear homogeneous equation with constant coefficients. We can find the characteristic equation

$$\lambda^2 + \lambda - 2 = 0.$$

Solving this (either by factorising or using the quadratic formula if necessary gives $\lambda_1 = 1$ and $\lambda_2 = -2$. The general solution will then be:

$$y = c_1 e^x + c_2 e^{-2x}$$

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Case 2: One real root (ie $a^2 - 4b = 0$):

In this case there is only one value for λ (even though it 'occurs twice'). We have one solution $y = e^{\lambda x}$, but need another. In this case it is also true that $y = xe^{\lambda x}$ is a solution (check!).

Our two solutions are linearly independent (they are not proportional). So a general solution is

$$y = c_1 e^{\lambda x} + c_2 x e^{\lambda x}$$

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Exercise: Find the general solution to the differential equation

$$y'' + 2y' + y = 0$$

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Case 3: Complex conjugate roots (ie $a^2 - 4b < 0$):
In this case there are no *real* roots to the characteristic equation, but we will get two complex roots. After some manipulation (see the last part of chapter 2.2 in the text book for details), we can obtain linearly independent solutions

$$y_1 = e^{-\frac{ax}{2}} \cos(\omega x) \quad \text{and} \quad y_2 = e^{-\frac{ax}{2}} \sin(\omega x)$$

where

$$\omega^2 = b - \frac{1}{4}a^2.$$

Our general solution then becomes:

$$y = e^{-\frac{ax}{2}} \left(A \cos(\omega x) + B \sin(\omega x)\right)$$

Note that the complex solutions of the characteristic equation are given by $\lambda = -\frac{a}{2} \pm i\omega$

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The cases can be summarised as follows: To solve the the differential equation

$$y'' + ay' + by = 0,$$

find roots of the characteristic equation

$$\lambda^2 + a\lambda + b = 0.$$

The solutions will be as follows, and depend on the types of roots found.

Case	Roots	General Solution
1	Distinct real λ_1, λ_2	$y = c_1 e^{\lambda_1 x} + c_2 e^{\lambda_2 x}$
2	One real λ	$y = (c_1 + c_2 x)e^{\lambda x}$
3	Complex conjugate $\lambda_1 = -\frac{1}{2}a + i\omega$ $\lambda_2 = -\frac{1}{2}a - i\omega$	$y = e^{-\frac{ax}{2}}\left(A\cos(\omega x) + B\sin(\omega x)\right)$

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Example: Find the general solution to the differential equation

$$y'' + 2y' + 5y = 0$$

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Example: Find the general solution to the differential equation

$$y'' + 2y' + 5y = 0$$

In this case the characteristic equation has roots $-1 \pm 2i$. So $\omega = 2$, and our solution is

$$y = e^{-x}\left(A\cos(2x) + B\sin(2x)\right)$$

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Exercises: Find the general solutions to the following differential equations:

- $y'' + 4y' + 3y = 0$
- $y'' + 4y' + 4y = 0$
- $y'' + 4y' + 5y = 0$

Text Book References

Chapter 2.2